

Recall

1. Schauder Basis
2. Separability
3. Compact sets
4. Characterisation of compact sets
in finite dimensional NLS. (Need to complete)
5. Riesz lemma (Proof today)
6. Baire Category (")

X be a metric space &

$$A \subseteq X$$

- A is nowhere dense iff

$$\bar{A}^\circ = \emptyset$$

- Dense iff $\bar{A} = X$

- X is of first category

if $X = \bigcup_{n=1}^{\infty} U_n$, U_n nowhere dense subsets,

$$U_n \subseteq X.$$

- If X is not of first

category then it is 2nd category.

Remark: (X, d)

$$\bullet \quad X = \bigcup_{n=1}^{\infty} A_n \Rightarrow X = \bigcup_{n=1}^{\infty} \overline{A_n}$$

Lemma: Let (X, d) be a metric space, $A \subseteq X$.

Then A is closed and nowhere dense iff A^c is open & dense in X .

Proof: Suppose A is closed & nowhere dense.

Then,

$$\bar{A}^{\circ} = A^{\circ} = \emptyset.$$

Since A is closed $\Rightarrow A^c$ is open. ($\star$)

Claim: A^c is dense.

Let $x \in X$ be arbitrary.

Since $A^{\circ} = \emptyset$ for each $\epsilon > 0$

We must have $B(x, \epsilon) \cap A^c \neq \emptyset$.

$\therefore A^c$ is dense. ($\star$)

Now suppose A^c is open & dense.

Claim : $A^\circ = \emptyset$.

A^c is open $\Rightarrow A$ is closed.

Let $a \in A^\circ$.

Then $\exists \epsilon > 0$ s.t. $B(a, \epsilon) \cap A^c = \emptyset$

(that A^c is dense) $\Downarrow$

$\therefore A^\circ = \emptyset$.

Cor. (X, d) metric space.

X is of first category $\iff \{U_n\}$ open, dense subsets of X

$$\text{s.t. } \bigcap_{n=1}^{\infty} U_n = \emptyset$$

Proof: Let X is of first category.

Then $X = \bigcup_{n=1}^{\infty} V_n$ - V_n nowhere dense

$$\Rightarrow X = \bigcup_{n=1}^{\infty} \overline{V_n} \quad \leftarrow$$

$$\Rightarrow X^c = \bigcap_{n=1}^{\infty} \overline{V_n}^c$$

$$\Rightarrow \bigcap_{n=1}^{\infty} \overline{V_n}^c = \emptyset$$

So our choice of $U_n = \overline{V_n}^c$

2. U_n is open

• U_n is dense since

$\overline{V_n}$ is nowhere dense.

Converse.

— Please complete

Baire Category theo.

Let (X, d) be a complete metric space.

Suppose $(U_n)_{n \geq 1}$ is a seq of open dense set of X

Then $\bigcap_{n=1}^{\infty} U_n$ is dense in X .

Hence X is 2nd category.

Proof: To show that $\bigcap_{n=1}^{\infty} U_n$ is dense in X
we have to show for any $\epsilon > 0$ for any $x \in X$

$$B(x, \epsilon) \cap \left(\bigcap_{n=1}^{\infty} U_n \right) \neq \emptyset.$$

Observe: $y \in X, \quad r > 0$

then $0 < r' < r$

$$B[y, r'] \subseteq B(y, r).$$

$$\text{Let } r_1 = \epsilon/2$$

Since U_1 is dense in X

$$\exists a_1 \in U_1 \text{ s.t. } d(a_1, x) < r_1$$

Again U_1 is open then by using the previous argument

$$\exists 0 < r_2 < \epsilon/4 < r_1 \text{ s.t.}$$

$$\underline{B[a_1, r_2] \subseteq U_1}$$

• Now U_2 is dense in X .

$$\exists \underline{a_2 \in U_2} \text{ s.t.}$$

$$\underline{d(a_2, a_1) < r_2}$$

Hence $\underline{a_2 \in B(a_1, r_2)}$

$$\therefore a_2 \in U_2 \cap B(a_1, r_2)$$

Again since $U_2 \cap B(a_1, r_2)$ is
open

$$\therefore \exists r_3 \text{ s.t. } 0 < r_3 < \frac{\epsilon}{2^3} < r_2$$

$$\text{s.t. } \underline{B[a_2, r_3] \subseteq U_2 \cap B(a_1, r_2)}$$

Continuing this way we get ,
for each $n \in \mathbb{N}$, $\exists a_n \in U_n \cap B(a_{n-1}, r_n)$
s.t. $0 < r_{n+1} < \frac{\epsilon}{2^{n+1}}$ s.t.

$$d(a_n, a_{n-1}) < r_n \quad \& \quad .$$

$$B[a_n, r_{n+1}] \subseteq U_n \cap B(a_{n-1}, r_n)$$

Let

$$F_n = B[a_n, r_{n+1}]$$

• F_n 's are closed

$$F_{n+1} \subseteq F_n$$

$$\text{diam}(F_n) \leq 2r_{n+1}$$

$$\bullet \text{diam}(F_n) \rightarrow 0$$

$$\text{as } n \rightarrow \infty$$

So by Cantor's Intersection

theorem

$$\bigcap_{n=1}^{\infty} F_n \neq \emptyset$$

Let $y \in \bigcap_{n=1}^{\infty} F_n$.

Claim: $y \in \bigcap_{n=1}^{\infty} U_n$ & $d(x, y) < \epsilon$

$$F_n \subseteq U_n \quad \forall n.$$

$$\Rightarrow y \in \bigcap_{n=1}^{\infty} U_n$$

Since $y \in \bigcap F_n \Rightarrow y \in F_1$
 $= B[a_1, r_2]$

$$\therefore d(y, a_1) < r_2 = \frac{\epsilon}{4}.$$

$$\begin{aligned} \therefore d(x, y) &\leq \underbrace{d(x, a_1)} + d(a_1, y) \\ &< r_1 + r_1 < \epsilon. \end{aligned}$$

$$\therefore d(x, y) < \epsilon.$$

Since x was arbitrary
So $\bigcap_{n=1}^{\infty} U_n$ is dense.

Riesz lemma

$$X_0 \subsetneq X$$

proper
closed subspace

Then for each $0 < r < 1$

$\exists x_n \in X$ s.t. $\|x_n\| = 1$

$$\underline{r \leq \text{dist}(x_n, X_0) \leq 1.}$$

Proof:

Let

$x \in X - X_0$, X_0 is closed.

$$\text{dist}(x, X_0) = d.$$

$$\inf \{ \|x - u\|, u \in X_0 \}.$$

Since $x \notin X_0 = \overline{X_0}$

$$\therefore \text{dist}(x, X_0) \neq 0.$$

ie,

$$d > 0$$

Choose any r s.t

$$0 < r < 1$$

with

$$\frac{d}{r} > d.$$

$$= \inf \{ \|x - u\| \mid u \in X_0 \}$$

$\exists x_0 \in X_0$ s.t

$$\|x - x_0\| < \frac{d}{r}.$$

Define $x_r = \frac{x - x_0}{\|x - x_0\|}$

$$\|x_r\| = 1$$

$$\therefore \text{dist}(x_r, x_0)$$

$$= \text{dist}\left(\frac{x - x_0}{\|x - x_0\|}, x_0\right)$$

$$= \frac{1}{\|x - x_0\|} \text{dist}(x - x_0, x_0)$$

$$= \frac{1}{\|x - x_0\|} \text{dist}(x, x_0)$$

$$\geq \frac{r}{d} \cdot d = r$$

$$\therefore \text{dist}(x_n, X_0) \geq r \quad \text{--- (1)}$$

Now,

$$\begin{aligned} \text{dist}(x_n, X_0) &= \inf \{ \|x_n - u\| \mid u \in X_0 \} \\ &\leq \|x_n\| \quad (\because 0 \in X_0) \\ &= 1. \end{aligned}$$

$$\therefore \text{dist}(x_n, X_0) \leq 1. \quad \text{--- (2)}$$

$$\boxed{r \leq \text{dist}(x_r, x_0) \leq 1.}$$

$$\|x_r\| = 1.$$

Equivalent statements.

1. X is finite dimensional.
2. Every closed & bdd subsets of X is compact.
3. The closed unit ball

$D = \{x \in X \mid \|x\| \leq 1\}$ is compact

4. For some $x_0 \in X$
& $r > 0$, $B(x_0, r)$ is
totally bdd.

$(3) \Rightarrow (4)$

$(4) \Rightarrow (1)$.

Let $B(x_0, r)$ be t.b for
some $x_0 \in X$ & $r > 0$.

$$x_0 + \left(\frac{r}{2}\right) \{x \in X \mid \|x\|=1\} \subseteq B(x_0, r)$$

Since $B(x_0, r)$ is totally
bdd

$$\Rightarrow \underline{S} = \{x \in X \mid \|x\|=1\}$$

is totally bdd.

Let X be infinite dimensional and

$E = \{u_1, \dots, u_n, \dots\}$
is a den. basis.

Let $X_n = \text{span}\{u_1, \dots, u_n\}$

$\therefore X_n$ is closed
& $X_n \subset X$.

Also $X_n \subsetneq X_{n+1} \quad \forall n.$

By Riesz lemma,

$\exists x_{n+1} \in X_{n+1}$ s.t

$$\|x_{n+1}\| = 1$$

$$\text{dist}(x_{n+1}, X_n) > \frac{1}{2} \quad \forall n.$$

~~def~~

Claim $\|x_m - x_n\| \geq \frac{1}{2}$

$$\forall n \neq m.$$

$$\text{if } n > m$$

$$X_m \subseteq X_n$$

$$\|x_m - x_n\| \geq \frac{1}{2}$$

$$\forall n \neq m$$

$\therefore (x_n)$ does not have
Cauchy subseq.

$\therefore S$ is not totally
bald.
 $\Downarrow$

$\therefore X$ is finite dimensional.

